

PACE Solver Description Uni-Bremen-LB: Maximum Agreement Forest Problem

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Abstract

This document provides a brief overview of our submission to the lower-bound track of the PACE Challenge 2026 on the MAXIMUM AGREEMENT FOREST problem. Our solver first preprocesses the instance by utilizing a reduction rule to simplify the instance. It then computes a fast approximation to obtain a heuristic solution, and derives a MAXSAT formulation to compute a strong lower bound. The instance is subsequently solved using a branching algorithm, guided by the previously computed bounds.

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Supplementary Material Public Repository: https://github.com/pace26unibremen/PACE2026_LB

1 Introduction

This year's PACE challenge features the MAXIMUM AGREEMENT FOREST problem, a classical NP-hard problem in phylogenetics.¹ Given a set of phylogenetic trees $T_1, \dots, T_k$ on the same leaf-label set L , the MAXIMUM AGREEMENT FOREST problem asks for an agreement forest F with a minimum number of trees. An agreement forest for $T_1, \dots, T_k$ is a forest F that can be obtained from each T_i by removing the same number of edges and subsequently contracting any vertices not in L that have in- and out-degree at most one [3].

¹ <https://pacechallenge.org/2026/>

The solver described in the following was submitted to the lower-bound track of the PACE Challenge, which permits suboptimal solutions within an instance-specific threshold. It consists of three distinct steps: preprocessing using reduction rules, computing an upper and a lower bound, and solving the preprocessed instance with a branching algorithm that utilizes the precomputed bounds. The preprocessing step, described in detail in Section 2, exhaustively applies the subtree reduction rule to reduce the instance’s complexity. The reduced instance is then solved using multiple approximation algorithms, yielding both an upper bound and a first lower bound on the size of an optimal agreement forest. This approximation algorithm is described in detail in Section 3.

Subsequently, the solver alternates between improving the lower bound and the upper bound until both lie within the instance-specific threshold. To improve the lower bound, it relies on an incremental MAXSAT approach: the instance is translated into a hitting set problem, encoded as a MAXSAT instance, and solved incrementally using the **UWrMaxSat solver** [7]. This approach is further explained in Section 4. To improve the upper bound, the solver applies the branching algorithm described in Section 5, run with a limited time budget to iteratively refine the bound.

2 Subtree reduction rule

The subtree reduction rule [4] contracts each maximal subtree shared identically across all input trees into a single leaf. As such subtrees are necessarily preserved in any agreement forest, this reduces the instance’s size without affecting the optimal solution. This rule is well known in the literature [2, 4, 5].

3 Approximation

The approximation routines are only used to establish initial upper and lower bounds for the solution.

To obtain the upper bound, we use the linear-time 3-approximation algorithm proposed by Whidden and Zeh [10]. For the two-tree case, this algorithm constructs a valid agreement forest whose size is guaranteed to be at most three times the optimum.

The lower bound is obtained by computing certified bounds from feasible dual solutions within the duality-based approximation framework for MAF on two rooted binary trees [8, 6]. Initially the solver compute the inexpensive bound derived from the 3-approximation. If the 2-approximation finishes in a given time window, the lower bound is tightened.

4 MaxSAT Formulation

As mentioned in Section 1, in order to compute a lower bound the MAF problem is reduced to a HITTING SET problem. The reduction closely follows the ILP formulation introduced by Wu [11], which is based on the observation that an agreement forest for two binary trees T_1 and T_2 can be obtained by cutting a minimum number of edges in T_1 , subject to two types of constraints that ensure the resulting forest is also a valid forest for T_2 . Together, these constraints define a set of branches that must each be *hit* by at least one cut edge, yielding a HITTING SET formulation [11].

The HITTING SET problem naturally translates into a (partial) MAXSAT formulation [1], which is exploited to solve the ILP formulation described above. To this end, each constraint is translated into a hard clause, while each binary variable C_i is encoded as a unit soft clause

penalizing the cutting of edge e_i . Thus, the resulting MAXSAT solution directly yields a minimum-size set of edges to cut.

This MAXSAT instance is then solved incrementally using the **UWrMaxSat solver** [7], rather than being generated and solved as a whole. This is necessary because the naive formulation contains $\mathcal{O}(n^4)$ constraints, which does not scale to larger instances. The incremental approach has two key benefits: it produces a valid lower bound after every intermediate step, and it allows the solving process to be interrupted, enabling the solver to alternate between improving the lower and upper bounds, as described in Section 1, until both lie within the solution interval.

5 Branching

To improve the upper bound, the solver uses a branching algorithm. It recursively cuts edges in forests $F_1, \dots, F_k$ according to rules from the literature. The upper bound for branching solutions is initialized with the approximation from Section 3. The following rules are evaluated in order, and the first applicable one is applied. Except for the first rule, all consider a pair of terminals a, c in F_1 and compare their relationship across the remaining forests.

Single Vertex Tree Propagation

Trivially, if one forest $F_1, \dots, F_k$ contains a single vertex tree, any edges connected to leaves of the same label need to be cut in all other forests.

AC (Separate components)

If the pair a and c are siblings belong to different connected components in at least one forest, the rule branches into two cases: cut a , or cut c while protecting the edge incident to a [9, p.107, "Separate components (AC)"].

EqualPair

This reduction rule is applicable in the trivial case where a and c are siblings in all forests. Here, a and c get contracted into one representative terminal.

B (One pendant subtree)

This rule, adapted from [9, p.107, "One pendant subtree (B)"], applies if, in every forest, either (i) a and c are siblings, or (ii) for some leaves b_1, b_2 , one of a and c forms a sibling pair with b_i while the other is their uncle. In this case, the leaves b_i are cut in all forests.

2B (Two pendant subtrees and a common uncle)

This rule, adapted from [9, p.109, "Two pendant subtrees and a common uncle (2B)"], applies to leaves a, b_1, b_2, c and x if in every forest, either (i) a and c are siblings with uncle x , (ii) a is sibling to b_i , c their uncle, and x their great-uncle, or (iii) a is sibling to b_i , b_j their uncle, c their great-uncle, and x their great-great-uncle. Then, b_1 and b_2 are cut in all forests.

117 ABC (Multiple pendant subtrees)

118 If the sibling pair a, c belongs to the same connected component in every forest, the rule
 119 branches by cutting a , cutting c , or cutting all edges on the path between them. When
 120 cutting c , the edge incident to a is protected [9, p.109, "Multiple pendant subtrees (A3BC)].

121 6 Sketch of correctness

122 The solver never returns an infeasible agreement forest, and it never returns a solution
 123 whose size exceeds the accepted bound, as the MAXSAT formulation guarantees correct
 124 lower bounds. Feasibility follows from the way solutions are recorded: the branching search
 125 only stores a solution once the current state represents an agreement forest, and the initial
 126 incumbent produced by the approximation algorithm is an agreement forest as well. The
 127 correctness of the reduction and branching rules for two trees was proven in their respective
 128 sources.

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